MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY: 10545703

(Buffer Amplifier)

1. SCOPE

1.1 This specification covers the detail requirements for one type of circuit card assembly. The requirements of this specification together with the general requirements of specification MIL-C-50785 comprise the complete requirements for the Buffer Amplifier Circuit Card Assembly.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-50785	Circuit Card Assemblies, General Specification for

DRAWINGS

U.S. Army, Frankford Arsenal

10545425	Sight Current Generator
10545455	Receptacle
10545703	Circuit Card Assembly, Buffer Amplifier

PACKAGING DATA SHEET

10545703	Packaging of Circuit Card Assembly, Buffer Amplifier

(Copies of specifications, standards, drawings, and publications required by Suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 1285

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3. REQUIREMENTS

3.1 Item definition. This assembly is a single-ended printed circuit heard consisting of three operational amplifiers, two inverters, and a voltage reference-scaler. me assembly is-used to invert and/or amplify voltages within the Sight Current Generator Drawing Number 1054542S in which it is assembled.

3.1.1 Major component list. The components of this assembly shall be as specified on the Engineering Parts List (EPL] 10S45703.

3.1.2 *Government* furnished property list. Not applicable.

3.2 Characteristics

3.2.1 **Performance.** **The** assembly shall be capable of performing functions a. through f. when assembled in accordance with Drawing Number 10545703.

a, Operational amplifier [Al) shall amplify and invert one *of* the inputs required to develop the torque current voltage.

b. Operational amplifier (A21 shall amplify and invert one of the inputs required to develop the magnet current voltage.

c. Operational amplifier (A3) shall amplify and invert one *of* **the** inputs required to develop the magnet current voltage.

d. Inverter (Q7) shall invert one of the inputs required to control the "Ready to Fire" circuitry and "Acquisition Time Delay" circuitry.

e. Inverter (Q_8) shall invert one of the inputs required to control the "Ready to Fire" circuitry.

f. -Voltage reference scaler shall scale the +20 Vdc reference voltage to +1 Vdc reference voltage.

3.2.1.1 Electrical requirements. The offset voltage of each operational amplifier, with zero volts applied to the input pins, shall be zero Vdc ± 50 millivolts. The input and output pins are as specified in Table I.

Table I. Input and Output Pins

Amplifier	Input Pins	Output Pins
A_1	2	8
A_2	15	12
A_3	3	5

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3.2.1.2 Input and output levels

3.2.1.2.1 Input power. Input power shall be supplied to the assembly as specified in Table XI.

Table II. Input voltages

Pin	Voltage
4, 25	+15 Vdc :+2.0 percent
14	-15 Vdc ± 2.0 percent
1 and 28	Ground

3.2.1.2.2 Output levels. The requirements specified in 3.2.1,2.2.1 through 5.2.1.2,2.5 shall be met when supplied with the specified voltage input (Vin).

3.2.1.2.2.1 Pin 5 output. The output shall be [Vin (-1.0)] ± 0,5 percent between pin 5 and pin one with a Vin of -13 Vdc to +13 Vdc applied between pin 3 and pin one.

3.2.1.2.2.2 Pin 8 output. The output shall be [Vin (-0.8666)] ± 0.5 percent between pin 8 and pin one with a Vin of -15 Vdc to +13 Vdc applied between pin 2 and pin one.

3.2.1.2.2.3 Output Pin 12. lhe output shall be[Vin (-0.8666]]"S0.5 percent between pin 12 and pin one with aVin of -13 Vdc to +13 Vdc applied between pin IS and pin 28.

3.2.1.2.2.4 Output pin 20. the output, with a one K load between pin 20 and pin 28, shall be +12 Vdc ± l.0 percent between pin 20 and pin 28 with +20 Vdc input applied between pin 22 and pin 28.

3.2.1.2.2.5 Output pin 26. The output shall be +15 Vdc 21.0.percent between pin 26 and pin 28 with zero Vdc input applied between pin 18 and pin 28, and 0.5 20.S Vdc with +l5 Vdc input applied between pin 18 and pin 28.

3.2.1 .2.2.6 Output pin 27. The output shall be +15 Vdc 21.0 percent between pin 27 and pin 28 with zero Vdc input applied between pin 17 and pin 28, and 0.5 ± 0.5 Vdc with +15 Vdc input applied between pin 17 and pin 28.

3.2.1.3 Environmental. The assembly shall meet the environmental requirements of specification MIL-C-S078S.

3.2.2 Physical characteristics. The phYsical characteristics shall be as specified in drawing 10545703 and the engineering parts list.

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3,3 Design and construction

3.3.1 Production drawing. This assembly shall be fabricated in accordance with the requirements of specification MIL-C-S078S, drawing 10545703 and associated lists.

3.3.2 Standards of manufacture. The standards of manufacture shall be those specified in specification MIL-C-5078S.

3.3.3 Workmanship . The workmanship shall be as specified in specification MIL-C-50785

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4. QUALITY ASSURANCE PROVISION

4.1 General

4.1.1 Responsibility for inspection. The responsibility for inspection shall be as specified in specification MIL-C-S078S.

4.1.2 Special tests and examination. the assembly shall undergo first article and control testing in accordance with 4.1.2 of specification MIL-C-50785.

4.2 Quality conformance inspections. The assembly shall be inspected and tested in accordance with 4.2 of specification MIL-C-S078S,

4.2.1 Test equipment. (See Figure 1.)

Power supply 1 (PS 1) milliamperes 100 mv peak-to-peak,	The power supply shall be 15 ± 0.1 Vdc, 100 ripple less than 10 millivolts (rev) rms or
Powersupply (PS 2) milliamperes 100 mv peak-to-peak.	The powersupply shall be -15 ± 0.1 Vdc, 100 ripple less than 10 millivolts (rev) rms or

4.2.1.3 Power supply (PS3). The power supply shall be adjustable from -13 Vdc to +20 Vdc with accuracy of adjustment and regulation to ±.01 percent, 10 ma or greater.

4.2.1.4 Resistor. One resistor is required with a rating of 1000 ohms ± 5 percent..

4.2.1.5 Digital voltmeter (DMI). The Digital Voltmeter shall have an accuracy and resolution of .01 percent or better from 0 to 100 Vdc. Input impedance shall be greater than 10 megohms on all voltage scales.

4.2.2 Functional test

4.2.2.1 Test connections. Power supplies PS1 and PS2 and the resistor shall be connected to the assembly under test in accordance with Figure 1 using receptable 10545455.

4.2.2.2 Test procedure. Compliance with the requirements of 3.2.1.1 through 3.2.1.2.2.6 shall be verified as follows: Power supply (PS3) voltage shall be adjusted and connected to the specified input pin per Table III. The digital voltmeter shall be used to measure the voltage at the applicable output pin to the limits indicated in Table III.

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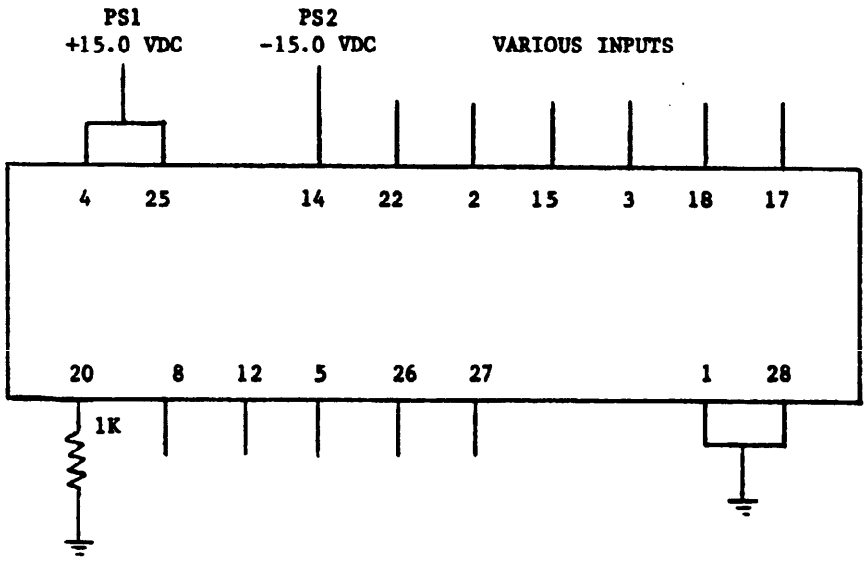


Figure 1. Test Connections

Table III. Test Limits

Input Voltage (Vdc)	Input Pin	Output Voltage (Vdc)	Output Pin
-13.0 ±0.013	2	+11.266 ±0.055	8
-13.0 ±0.013	15	+11.266 ±0.055	12
-13.0 ±0.013	3	+13.000 ±0.065	5
0 ±0.100	18	+15.000 ±0.150	26
0 ±0.100	17	+15.000 ±0.150	27
0 ±0.010	2	0 ±0.050	8
0 ±0.010	15	0 ±0.050	12
0 ±0.010	3	0 ±0.050	5
+13.0 ±0.013	2	-11.266 ±0.055	8
+13.0 ±0.013	15	-11.266 ±0.055	12
+13.0 ±0.013	3	-13.000 ±0.065	5
+15.0 ±0.10	18	+0.500 ±0.500	26
+15.0 ±0.10	17	+0.500 ±0.500	27
+20.0 ±0.03	22	+12.000 ±0.120	20

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5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging and marking shall be in accordance with the level A requirements of Packaging Data Sheet (PDS), AMC form 1029, 10545703.

6. NOTES

6.1 The notes specified in the general specification MIL-C-S0785 are applicable to this specification.

Custodian	Preparing Activity
Army - Mu	Army - Mu Project No. 1285-A313

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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